

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 03 · Circles

Questions

Student book · questions with writing space

- 57 pages
- Past papers up to January 2026

PURE 2 | TOPIC 03

Circles

Your classified route

19

QUESTIONS

154

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 03 | FAMILY 01

Circle Equation

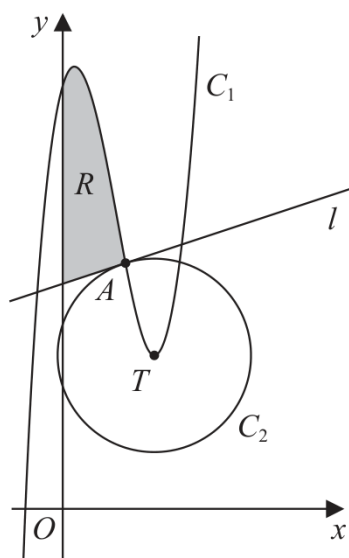
WHAT THIS FAMILY ASKS

Recognise the circle equation demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**3 classified questions**

26 total marks

9.

In this question you must show detailed reasoning.**Solutions relying entirely on calculator technology are not acceptable.****Figure 3**

Adaptive unit 1 · WMA12-01_2022_Oct_Q09

Figure 3 shows

- the curve C_1 with equation $y = x^3 - 5x^2 + 3x + 14$
- the circle C_2 with centre T

The point T is the minimum turning point of C_1

Using Figure 3 and calculus,

(a) find the coordinates of T

(3)

Adaptive unit 1 · WMA12-01_2022_Oct_Q09

DR ESAM AHMED

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

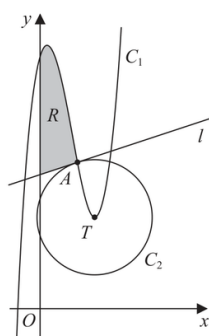


Figure 3

Figure 3 shows

- the curve C_1 with equation $y = x^3 - 5x^2 + 3x + 14$
- the circle C_2 with centre T

The point T is the minimum turning point of C_1

Using Figure 3 and calculus,

The curve C_1 intersects the circle C_2 at the point A with x coordinate 2(b) Find an equation of the circle C_2

(3)

Adaptive unit 2 · WMA12-01_2022_Oct_Q09

9. In this question you must show detailed reasoning.
Solutions relying entirely on calculator technology are not acceptable.

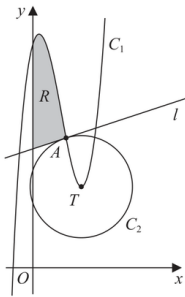


Figure 3

- Figure 3 shows
- the curve C_1 with equation $y = x^3 - 5x^2 + 3x + 14$
 - the circle C_2 with centre T
- The point T is the minimum turning point of C_1
- Using Figure 3 and calculus,

The line l shown in Figure 3, is the tangent to circle C_2 at A

(c) Show that an equation of l is

$$y = \frac{1}{3}x + \frac{22}{3}$$

(3)

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

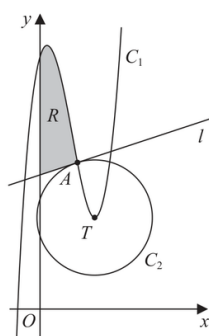


Figure 3

Figure 3 shows

- the curve C_1 with equation $y = x^3 - 5x^2 + 3x + 14$
- the circle C_2 with centre T

The point T is the minimum turning point of C_1

Using Figure 3 and calculus,

The region R , shown shaded in Figure 3, is bounded by C_1 , l and the y -axis.(d) Find the exact area of R .

(3)

Adaptive unit 4 · WMA12-01_2022_Oct_Q09

3. The circle C

- has centre $A(3, 5)$
 - passes through the point $B(8, -7)$
- (a) Find an equation for C .

(3)

Adaptive unit 1 · WMA12-01_2024_Jan_Q03

DR ESAM AHMED

3. The circle C

- has centre $A(3, 5)$
- passes through the point $B(8, -7)$

The points M and N lie on C such that MN is a chord of C .

Given that MN

- lies above the x -axis
- is parallel to the x -axis
- has length $4\sqrt{22}$

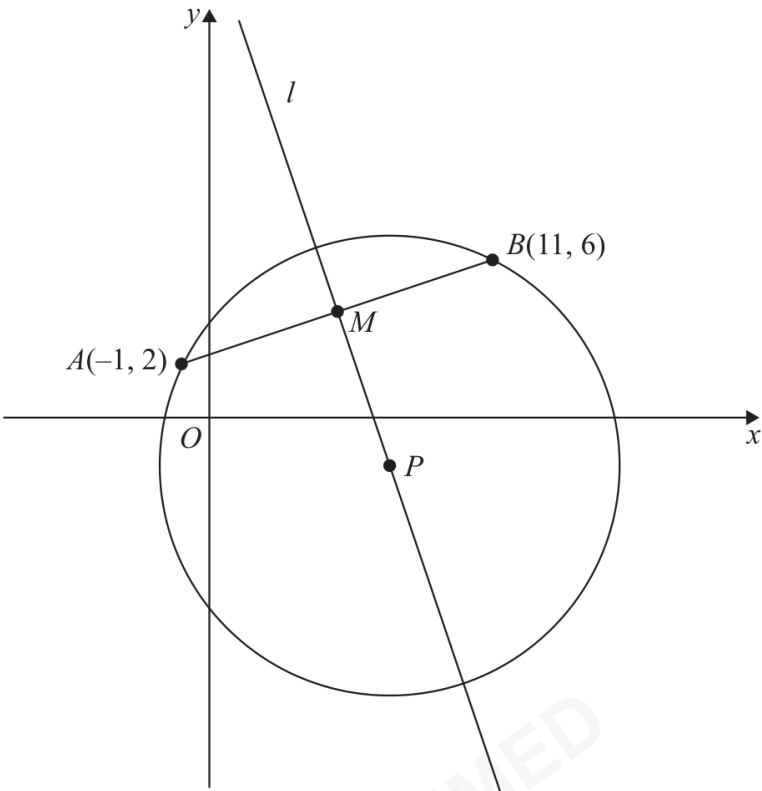
(b) find an equation for the line passing through points M and N .

(3)

Adaptive unit 2 · WMA12-01_2024_Jan_Q03

DR ESAM AHMED

6.



Adaptive unit 1 · WMA12-01_2025_Jan_Q06

Figure 2

The point $A (-1, 2)$ and the point $B (11, 6)$ both lie on a circle with centre P .

The point M is the midpoint of AB .

Given that the line l passes through M and P , as shown in Figure 2,

- (a) find an equation for l , giving your answer in the form $y = mx + c$, where m and c are constants.

(4)

Adaptive unit 1 · WMA12-01_2025_Jan_Q06

DR ESAM AHMED

6.

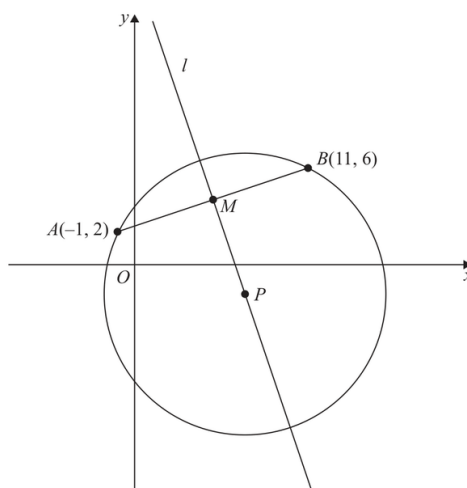


Figure 2

The point $A(-1, 2)$ and the point $B(11, 6)$ both lie on a circle with centre P .

The point M is the midpoint of AB .

Given that the line l passes through M and P , as shown in Figure 2,

Given that P has coordinates $(7, k)$, where k is a constant,

(b) find the value of k ,

(1)

Adaptive unit 2 · WMA12-01_2025_Jan_Q06

6.

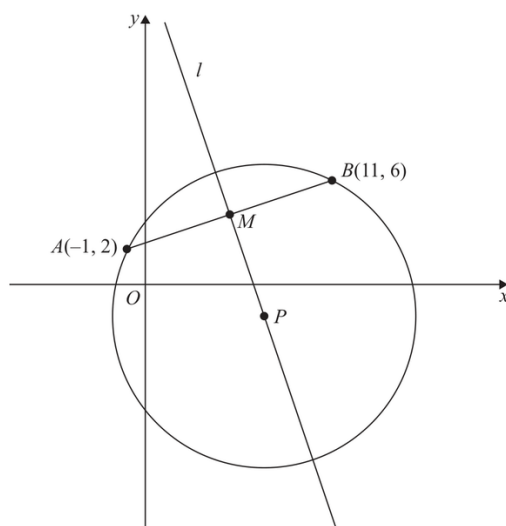


Figure 2

The point $A(-1, 2)$ and the point $B(11, 6)$ both lie on a circle with centre P .

The point M is the midpoint of AB .

Given that the line l passes through M and P , as shown in Figure 2,

(c) find an equation for the circle.

(3)

Adaptive unit 3 · WMA12-01_2025_Jan_Q06

TOPIC 03 | FAMILY 02

Line-Circle Intersections

WHAT THIS FAMILY ASKS

Recognise the line-circle intersections demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 Tangency means the quadratic has one repeated root.

1 classified questions

12 total marks

10. The circle C has centre $X(3, 5)$ and radius r

The line l has equation $y = 2x + k$, where k is a constant.

(a) Show that l and C intersect when

$$5x^2 + (4k - 26)x + k^2 - 10k + 34 - r^2 = 0$$

(3)

Adaptive unit 1 · WMA12-01_2022_MayJune_Q10

DR ESLAM AHMED

10. The circle C has centre $X(3, 5)$ and radius r
The line l has equation $y = 2x + k$, where k is a constant.

Given that l is a tangent to C ,
(b) show that $5r^2 = (k + p)^2$, where p is a constant to be found.

(3)

Adaptive unit 2 · WMA12-01_2022_MayJune_Q10

DR ESAM AHMED

10. The circle C has centre $X(3, 5)$ and radius r

The line l has equation $y = 2x + k$, where k is a constant.

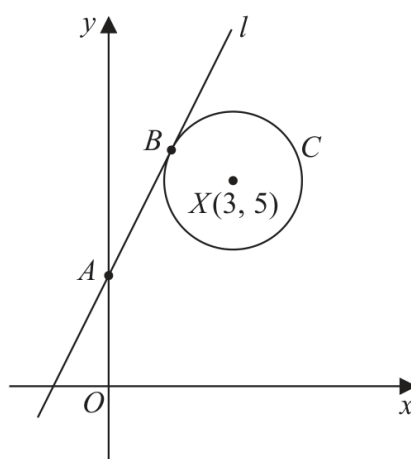


Figure 2

The line l

- cuts the y -axis at the point A
- touches the circle C at the point B

Adaptive unit 3 · WMA12-01_2022_MayJune_Q10

as shown in Figure 2.

Given that $AB = 2r$

(c) find the value of k

(6)

Adaptive unit 3 · WMA12-01_2022_MayJune_Q10

DR ESAM AHMED

TOPIC 03 | FAMILY 03

Radius/Distance

WHAT THIS FAMILY ASKS

Recognise the radius/distance demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**7 classified questions**

49 total marks

6. The circle C has equation

$$x^2 + y^2 + 6x - 4y - 14 = 0$$

- (a) Find
- (i) the coordinates of the centre of C ,
 - (ii) the exact radius of C .

(3)

Adaptive unit 1 · WMA12-01_2020_Jan_Q06

DR ESAM AHMED

6. The circle C has equation

$$x^2 + y^2 + 6x - 4y - 14 = 0$$

The line with equation $y = k$, where k is a constant, is a tangent to C .

(b) Find the possible values of k .

(2)

Adaptive unit 2 · WMA12-01_2020_Jan_Q06

DR ESAM AHMED

6. The circle C has equation

$$x^2 + y^2 + 6x - 4y - 14 = 0$$

The line with equation $y = p$, where p is a negative constant, is a chord of C .

Given that the length of this chord is 4 units,

(c) find the value of p .

(3)

Adaptive unit 3 · WMA12-01_2020_Jan_Q06

DR ESAM AHMED

6. (i) The circle C_1 has equation

$$x^2 + y^2 + 10x - 12y = k \qquad \text{where } k \text{ is a constant}$$

(a) Find the coordinates of the centre of C_1 (2)

(b) State the possible range in values for k . (2)

Adaptive unit 1 · WMA12-01_2021_Oct_Q06

DR ESAM AHMED

- (ii) The point $P(p, 0)$, the point $Q(-2, 10)$ and the point $R(8, -14)$ lie on a different circle, C_2

Given that

- p is a positive constant
- QR is a diameter of C_2

find the exact value of p .

(4)

Adaptive unit 2 · WMA12-01_2021_Oct_Q06

DR ESAM AHMED

7. The circle C_1 has equation

$$x^2 + y^2 + 8x - 10y = 29$$

- (a) (i) Find the coordinates of the centre of C_1
(ii) Find the exact value of the radius of C_1

(3)

Adaptive unit 1 · WMA12-01_2024_MayJune_Q07

DR ESAM AHMED

7. The circle C_1 has equation

$$x^2 + y^2 + 8x - 10y = 29$$

In part (b) you must show detailed reasoning.

The circle C_2 has equation

$$(x - 5)^2 + (y + 8)^2 = 52$$

(b) Prove that the circles C_1 and C_2 neither touch nor intersect.

(3)

Adaptive unit 2 · WMA12-01_2024_MayJune_Q07

DR ESAM AHMED

3. A circle has equation

$$x^2 + y^2 + 8x - 14y - 79 = 0$$

- (a) Find
- (i) the coordinates of the centre of the circle,
 - (ii) the radius of the circle.

(3)

Adaptive unit 1 · WMA12-01_2024_MayJuneR_Q03

DR ESAM AHMED

3. A circle has equation

$$x^2 + y^2 + 8x - 14y - 79 = 0$$

Given that P is the point on the circle that is nearest the origin O ,

(b) find the exact length of OP

(2)

Adaptive unit 2 · WMA12-01_2024_MayJuneR_Q03

DR ESAM AHMED

- 2: The line joining the points $(-2, 5)$ and $(4, 15)$ is the diameter of a circle C .
- (a) Find an equation for C .

(5)

Adaptive unit 1 · WMA12-01_2025_MayJune_Q02

DR ESAM AHMED

2: The line joining the points $(-2, 5)$ and $(4, 15)$ is the diameter of a circle C .

(b) Hence find the exact coordinates of the point on C that is nearest the x -axis.

(2)

Adaptive unit 2 · WMA12-01_2025_MayJune_Q02

DR ESAM AHMED

6.

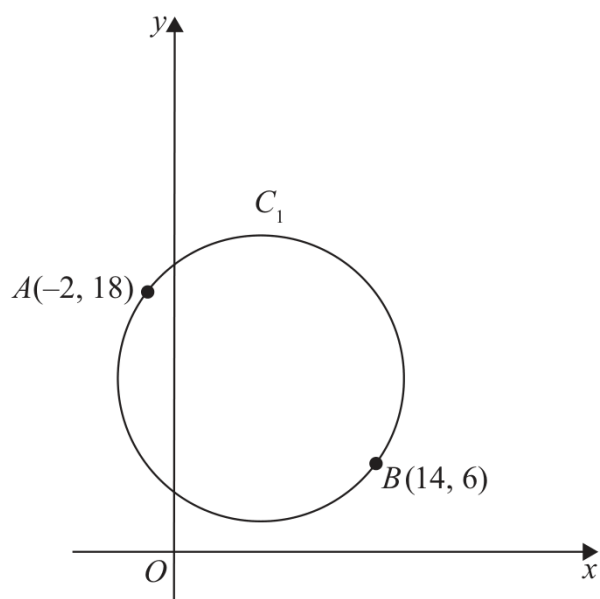


Figure 2

Adaptive unit 1 · WMA12-01_2025_Oct_Q06

The points $A(-2, 18)$ and $B(14, 6)$ lie on a circle C_1 as shown in Figure 2.

Given that AB is a diameter of the circle C_1

(a) find an equation for C_1 making your method clear.

(5)

Adaptive unit 1 · WMA12-01_2025_Oct_Q06

DR ESAM AHMED

6.

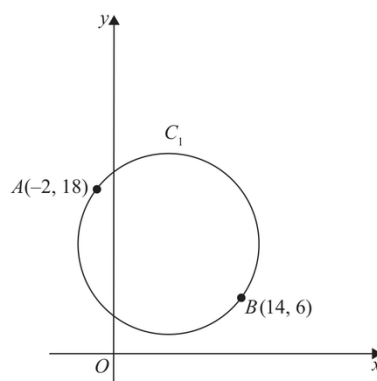


Figure 2

The points $A(-2, 18)$ and $B(14, 6)$ lie on a circle C_1 as shown in Figure 2.

Given that AB is a diameter of the circle C_1

A circle C_2 has its centre at the origin.

Given that circles C_1 and C_2 touch,

(b) find possible equations for C_2

(4)

Adaptive unit 2 · WMA12-01_2025_Oct_Q06

5. The circle C_1 has equation

$$x^2 + y^2 - 6x + 5y - 41 = 0$$

- (a) Find
- (i) the coordinates of the centre of C_1
 - (ii) the radius of C_1

(3)

Adaptive unit 1 · WMA12-01_2026_Jan_Q05

DR ESAM AHMED

5. The circle C_1 has equation

$$x^2 + y^2 - 6x + 5y - 41 = 0$$

The circle C_2 has

- centre $(-k, 0)$ where k is a positive constant
- radius 5

Given that circles C_1 and C_2 touch

(b) find the exact value of k .

(3)

Adaptive unit 2 · WMA12-01_2026_Jan_Q05

DR ESAM AHMED

TOPIC 03 | FAMILY 04

Tangents

WHAT THIS FAMILY ASKS

Recognise the tangents demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 If the radius is vertical, the tangent is horizontal.

8 classified questions

67 total marks

6.

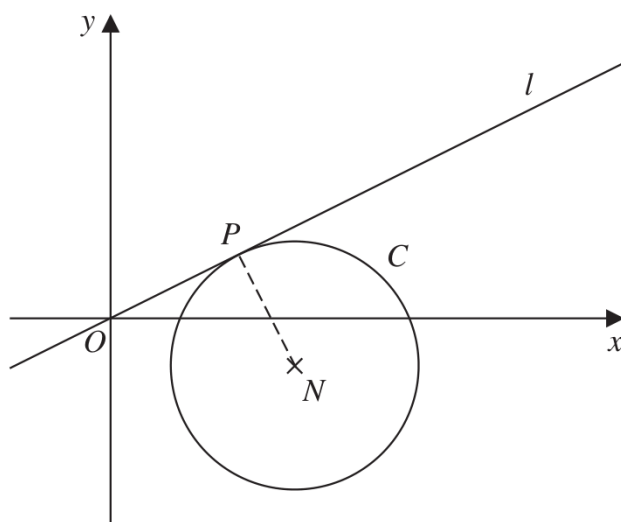
**Figure 1**

Figure 1 shows a sketch of a circle C with centre $N(4, -1)$.

Adaptive unit 1 · WMA12-01_2019_Oct_Q06

The line l with equation $y = \frac{1}{2}x$ is a tangent to C at the point P .

Find

- (a) the equation of line PN in the form $y = mx + c$, where m and c are constants, (2)

Adaptive unit 1 · WMA12-01_2019_Oct_Q06

DR ESAM AHMED

6.

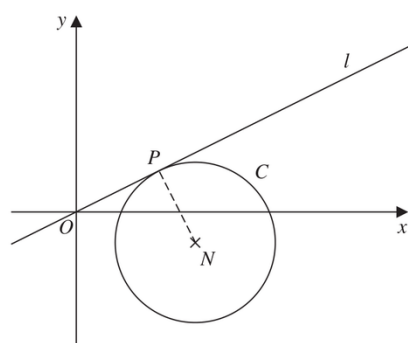
**Figure 1**

Figure 1 shows a sketch of a circle C with centre $N(4, -1)$.

The line l with equation $y = \frac{1}{2}x$ is a tangent to C at the point P .

Find

(b) the equation of C .

(5)

Adaptive unit 2 · WMA12-01_2019_Oct_Q06

DR ESAM AHMED

9. A circle C has equation

$$(x - k)^2 + (y - 2k)^2 = k + 7$$

where k is a positive constant.

- (a) Write down, in terms of k ,
- (i) the coordinates of the centre of C ,
 - (ii) the radius of C .

(2)

Adaptive unit 1 · WMA12-01_2021_Jan_Q09

DR ESAM AHMED

9. A circle C has equation

$$(x - k)^2 + (y - 2k)^2 = k + 7$$

where k is a positive constant.

Given that the point $P(2, 3)$ lies on C

(b) (i) show that $5k^2 - 17k + 6 = 0$

(ii) hence find the possible values of k .

(3)

Adaptive unit 2 · WMA12-01_2021_Jan_Q09

DR ESAM AHMED

9. A circle C has equation

$$(x - k)^2 + (y - 2k)^2 = k + 7$$

where k is a positive constant.

The tangent to the circle at P intersects the x -axis at point T .

Given that $k < 2$

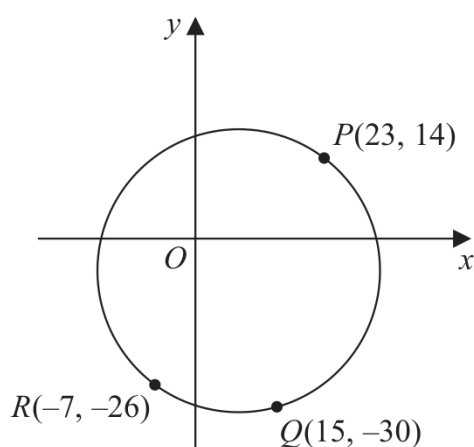
- (c) calculate the exact area of triangle OPT .

(5)

Adaptive unit 3 · WMA12-01_2021_Jan_Q09

DR ESAM AHMED

6.

**Figure 1**

The points $P(23, 14)$, $Q(15, -30)$ and $R(-7, -26)$ lie on the circle C , as shown in Figure 1.

(a) Show that angle $PQR = 90^\circ$

(2)

Adaptive unit 1 · WMA12-01_2022_Jan_Q06

6.

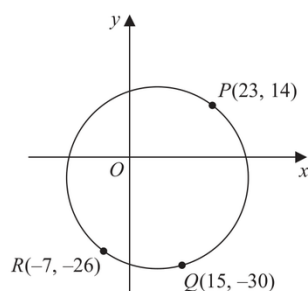


Figure 1

The points $P(23, 14)$, $Q(15, -30)$ and $R(-7, -26)$ lie on the circle C , as shown in Figure 1.

(b) Hence, or otherwise, find

- (i) the centre of C ,
- (ii) the radius of C .

(3)

Adaptive unit 2 · WMA12-01_2022_Jan_Q06

DR ESAM AHMED

6.

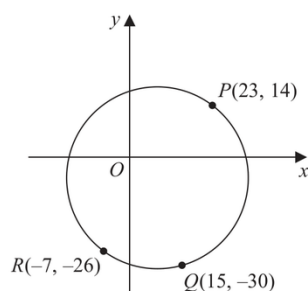


Figure 1

The points $P(23, 14)$, $Q(15, -30)$ and $R(-7, -26)$ lie on the circle C , as shown in Figure 1.

Given that the point S lies on C such that the distance QS is greatest,

- (c) find an equation of the tangent to C at S , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

Adaptive unit 3 · WMA12-01_2022_Jan_Q06

DR ESAM AHMED

6. The circle C has equation

$$x^2 + y^2 + 8x - 4y = 0$$

- (a) Find
- (i) the coordinates of the centre of C ,
 - (ii) the exact radius of C .

(3)

Adaptive unit 1 · WMA12-01_2023_Jan_Q06

DR ESAM AHMED

6. The circle C has equation

$$x^2 + y^2 + 8x - 4y = 0$$

The point P lies on C .

Given that the tangent to C at P has equation $x + 2y + 10 = 0$

(b) find the coordinates of P

(4)

Adaptive unit 2 · WMA12-01_2023_Jan_Q06

DR ESAM AHMED

6. The circle C has equation

$$x^2 + y^2 + 8x - 4y = 0$$

(c) Find the equation of the normal to C at P , giving your answer in the form $y = mx + c$ where m and c are integers to be found.

(3)

Adaptive unit 3 · WMA12-01_2023_Jan_Q06

DR ESAM AHMED

3. A circle C has centre $(2, 5)$
- Given that the point $P(8, -3)$ lies on C
- (a) (i) find the radius of C
- (ii) find an equation for C

(3)

Adaptive unit 1 · WMA12-01_2023_MayJune_Q03

DR ESAM AHMED

3. A circle C has centre $(2, 5)$
Given that the point $P(8, -3)$ lies on C

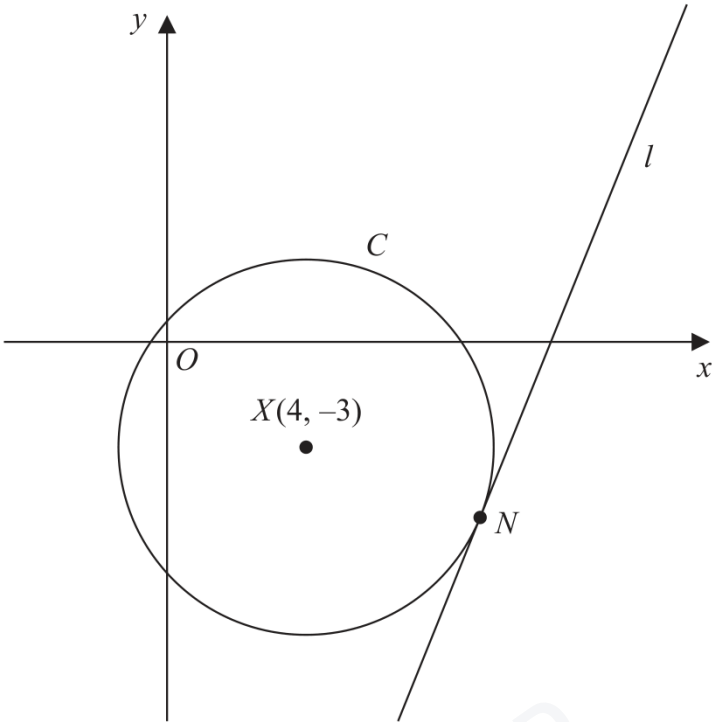
(b) Find the equation of the tangent to C at P giving your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found.

(4)

Adaptive unit 2 · WMA12-01_2023_MayJune_Q03

DR ESAM AHMED

7.



Adaptive unit 1 · WMA12-01_2023_Oct_Q07

DR ESAM AHMED

Figure 2

Figure 2 shows a sketch of

- the circle C with centre $X(4, -3)$
- the line l with equation $y = \frac{5}{2}x - \frac{55}{2}$

Given that l is the tangent to C at the point N ,

(a) show that an equation for the straight line passing through X and N is

$$2x + 5y + 7 = 0$$

(3)

Adaptive unit 1 · WMA12-01_2023_Oct_Q07

DR ESAM AHMED

7.

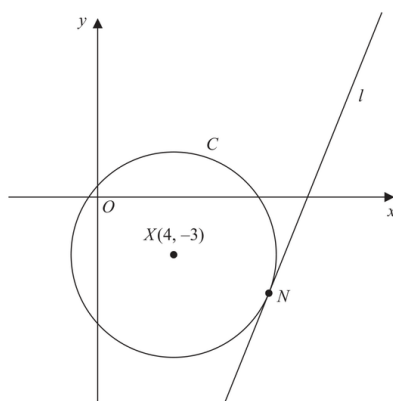


Figure 2

Figure 2 shows a sketch of

- the circle C with centre $X(4, -3)$
- the line l with equation $y = \frac{5}{2}x - \frac{55}{2}$

Given that l is the tangent to C at the point N ,

(b) Hence find

- the coordinates of N ,
- an equation for C .

(5)

Adaptive unit 2 · WMA12-01_2023_Oct_Q07

10. The circle C has equation

$$x^2 + y^2 + 4x - 30y + 209 = 0$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact value of the radius of C .

(3)

The line L has equation $y = mx + 1$, where m is a constant.

Given that L is the tangent to C at the point P ,

(b) show that

$$2m^2 - 7m - 22 = 0$$

(3)

Adaptive unit 1 · WMA12-01_2024_Oct_Q10

DR ESAM AHMED

10. The circle C has equation

$$x^2 + y^2 + 4x - 30y + 209 = 0$$

(c) Hence find the possible pairs of coordinates of P .

(4)

Adaptive unit 2 · WMA12-01_2024_Oct_Q10

DR ESAM AHMED

6. A circle has equation

$$x^2 - 6x + y^2 + 8y + k = 0$$

where k is a positive constant.

Given that the x -axis is a tangent to this circle,

- (a) find the value of k .

(3)

Adaptive unit 1 · WMA12-01_2021_MayJune_Q06

DR ESAM AHMED

6. A circle has equation

$$x^2 - 6x + y^2 + 8y + k = 0$$

where k is a positive constant.

Given that the x -axis is a tangent to this circle,

The circle meets the coordinate axes at the points R , S and T .

(b) Find the exact area of the triangle RST .

(4)

Adaptive unit 2 · WMA12-01_2021_MayJune_Q06

DR ESAM AHMED